χ^2 Tests

χ^2 Distribution

- A test of this sort was originally proposed by Karl Pearson in 1900.
- Let $X_i \sim N(\mu_i, \sigma_i^2)$ and $X_1, \ldots, X_n$ are mutually independent.

$$\sum_{i=1}^{n} \frac{(X_i - \mu_i)^2}{\sigma_i^2} \sim \chi^2(n)$$

Approximate χ^2 Distribution

- Let $X_1 \sim Bin(n, p_1)$. Consider the random variable

$$Y = \frac{X_1 - np_1}{\sqrt{np_1(1-p_1)}}.$$

  Y_1 is approximately $N(0,1)$ by the CLT, and Y_1^2 is approximately $\chi^2(1)$.

- Let $X_2 = n - X_1$ and let $p_2 = 1 - p_1$. Let $Q_1 = Y_1^2$.

$$Q_1 = \frac{(X_1 - np_1)^2}{np_1(1-p_1)} = \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2}$$

Approximate χ^2 Distribution

- Let $X_1, X_2, \ldots, X_{k-1} \sim multi(n; p_1, \ldots, p_k)$. Let $X_k = n - \sum_{i=1}^{k-1} X_i$ and $p_k = 1 - \sum_{i=1}^{k-1} p_i$. Define Q_{k-1} by

$$Q_{k-1} = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$$

- Q_{k-1} is approximately $\chi^2(k-1)$.

χ^2 Test

- Sample space $\mathcal{A} = \bigcup_{i=1}^{k} A_i$ and $\mathbb{P}(A_i) = p_i$, $\sum_{i=1}^{k} p_i = 1$
- Repeat the experiment n times independently. Let X_i be the number of times the outcome is an element of A_i

$$H_0 : p_1 = p_{10}, \ldots, p_k = p_{k0} \quad vs. \quad H_1 : \text{not } \mathrm{H}_0$$

- Test statistic:

$$Q_{k-1} := \sum_{i=1}^{k} \frac{(X_i - np_{i0})^2}{np_{i0}}$$

- When H_0 is true, $Q_{k-1} \dot{\sim} \chi^2_{k-1}$
- Decision rule: Reject H_0 if $Q_{k-1} \geq c$ and $c = \chi^2_{k-1,\alpha}$
- Goodness of fit test

Examples

- Cast a die. $A_i = \{x : x = i\}$, $i = 1, \ldots, 6$.

$$H_0 : p_{i0} = \frac{1}{6}, \ i = 1, \ldots, 6$$

- Data: $n = 60$. Observed counts $\{13, 19, 11, 8, 5, 4\}$
- Expected counts $\{10, 10, 10, 10, 10, 10\}$

$$Q_5 = \sum_{i=1}^{6} \frac{(x_i - 10)^2}{10}$$

- Under H_0, $Q_5 \dot{\sim} \chi^2_5$. Reject H_0 if $Q_5 \geq \chi^2_{5,0.05} = 11.1$. Oberve $Q_5 = 15.6$
- Reject H_0 at the (approximate) 5% significant level.

Examples

- $A_1 = (0, \frac{1}{4}]$, $A_2 = (\frac{1}{4}, \frac{1}{2}]$, $A_3 = (\frac{1}{2}, \frac{3}{4}]$, $A_4 = (\frac{3}{4}, 1)$.
- pdf: $f(x) = 2x$, $0 < x < 1$.

$$H_0 : p_{10} = \int_0^{1/4} 2x dx = \frac{7}{16}, p_{20} = \frac{3}{16}, p_{30} = \frac{5}{16}, p_{40} = \frac{7}{16}$$

- Data: $n = 80$. Observed counts: $\{6, 18, 20, 36\}$. Expected counts: $\{5, 16, 25, 35\}$
- When H_0 is true,

$$Q_3 = \sum_{i=1}^{4} \frac{(x_i - np_{i0})^2}{np_{i0}} \dot{\sim} \chi^2_3$$

  Observe $Q_3 = 64/35 = 1.83$.
- Reject H_0 if $Q_3 \geq \chi^2_{3,0.025} = 9.35$.
- Accept H_0 at the (approximate) 2.5% level of significance.

χ^2 test for Independence for Contingency Table

The hair colors and eye colors of Scottish children

	Fair	Red	Medium	Dark	Black	Margin
Blue	1368	170	1041	398	1	2798
Light	2577	474	2703	932	11	6697
Medium	1390	420	3286	1842	33	7511
Dark	454	255	1848	2506	112	5175
Margin	5789	1319	9418	5678	157	22361

Are hair and eye color independent?

χ^2 test for Independence for Contingency Table

Test:

$$H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}, i = 1, \ldots, a; j = 1, \ldots, b$$

Under H_0, the χ^2 test statistic has an approximate χ^2 distribution with $ab - 1 - (a + b - 2) = (a - 1)(b - 1)$ degrees of freedom.

If independent, the expected frequency with blue eyes and black hair is $2978 \times 157/22361 = 20.9$. The contribution to the test statistic from this one cell is $(1 - 20.9)^2/20.9 = 19.95$. χ^2 test statistic is 3683.9. p-value is $2.2e - 16$. $\chi^2_{0.95,12} = 21.026$. Reject H_0.

Based on this study, hair color and eye color of Scottish children are dependent on one another.